AI-Enabled Digital Twin for Early Prediction of Glucose Spikes

Team Codexxx | Chandigarh University

Happiest Health Digital Twin Challenge, Phase 1 prototype

The Problem



Meals

Carbohydrate intake drives post-meal glucose rises.



Physical activity

Exercise and inactive periods shift glucose levels.



Individual health

HbA1c, BMI and history change each patient's baseline.

2 h

early-warning window our twin aims to give

A current glucose reading alone gives no warning of a coming spike.

Our Solution: Four Stages

1 Generate

Synthetic EHR and wearable datasets

2 Fuse

Join patient profiles with time-series on patient ID

3 Predict

Random Forest estimates 2-hour spike risk

4 Display

Streamlit + Plotly dashboard for the doctor

Educational prototype on synthetic data. Not a clinically validated medical system.

Two Data Streams, Fused

Static EHR data

- Demographics, age, sex
- BMI
- HbA1c and fasting glucose
- Systolic blood pressure
- Diabetes and hypertension history
- Family history, genetic risk score
- Medication information

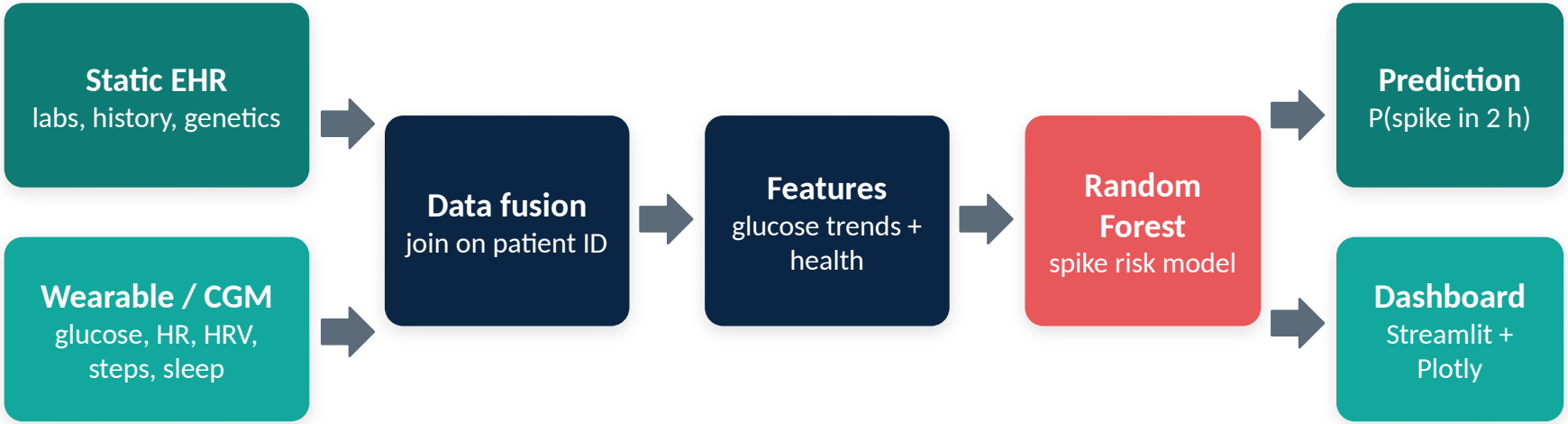
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Wearable / CGM data

- Glucose measurements (CGM)
- Heart rate
- Heart rate variability (HRV)
- Steps
- Sleep duration
- Time index

Target: glucose_spike_2h
1 = simulated glucose reaches 180 mg/dL or higher within the next 2 hours

System Architecture



Full diagram: docs/architecture.pdf in the repository

The Model: Random Forest



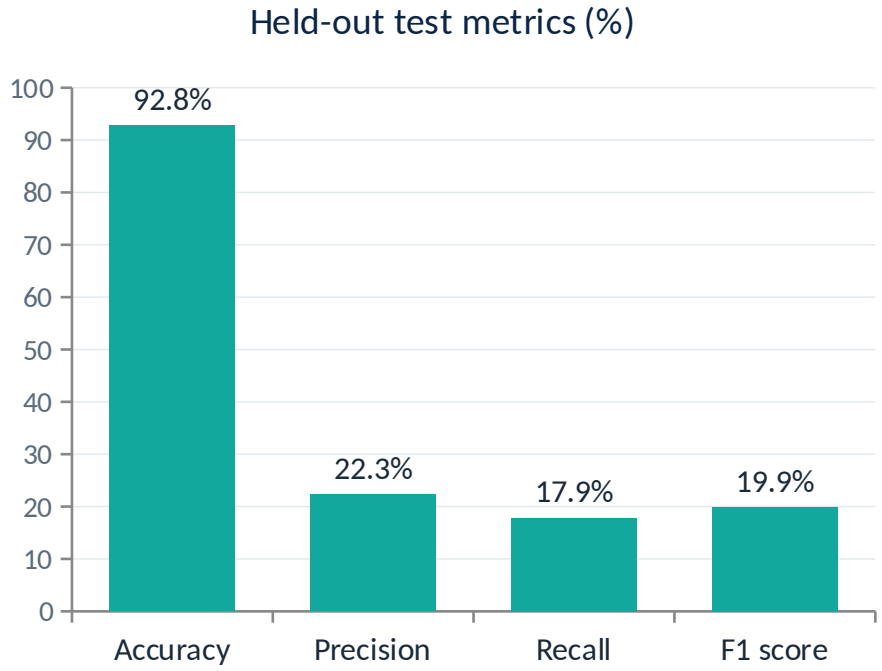
- Scikit-learn Random Forest classifier
- Input: fused EHR + wearable features
- Output: probability of a spike within 2 hours
- Gives feature importance for the dashboard
- Saved with Joblib as glucose_spike_model.pkl

17,600
samples in the held-out synthetic test set

879
spike cases in the test set (about 5%)

180
mg/dL threshold that defines a spike

Model Performance



Confusion matrix

	Predicted: no spike	Predicted: spike
Actual: no spike	**16,175** true negative	**546** false positive
Actual: spike	**722** false negative	**157** true positive

ROC-AUC **0.876**

Reading the Results Honestly

95.0%

accuracy of always predicting "no spike"

92.8%

accuracy of our Random Forest

Only about 5% of samples are spikes, so accuracy alone misleads. ROC-AUC (0.876) is the better measure here.

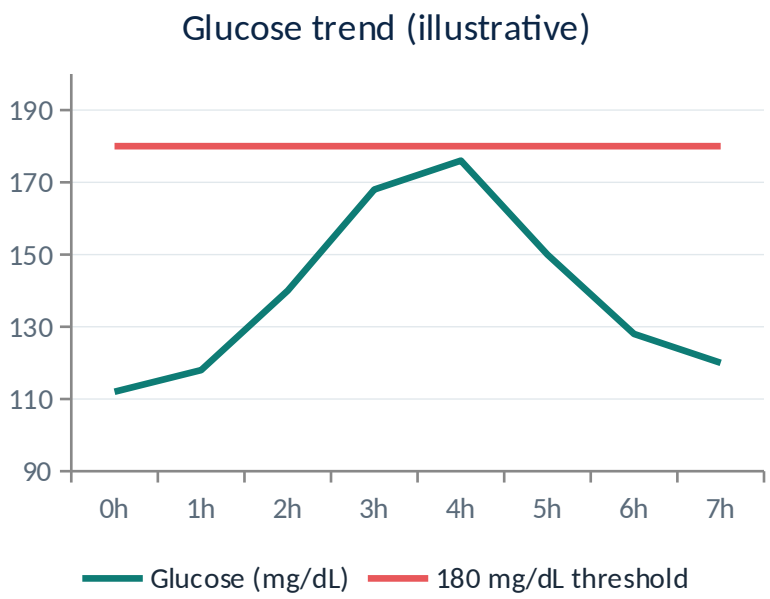
What we would improve next

- Class weighting, so rare spikes count more
- Threshold tuning below the default 0.5 to raise recall
- Richer time-series features such as rolling means and rate of change
- Patient-wise validation splits

The Doctor Dashboard

Patient profile

- Age and sex
- BMI
- HbA1c, fasting glucose
- Diabetes / BP history
- Genetic risk score
- Medication



2-hour spike risk

model probability, shown as a risk level

Feature importance

Layout is conceptual and the curve is illustrative. The working app is dashboard/app.py (screenshot in docs/dashboard.png).

Limitations and Future Scope

Limitations

- Trained and tested on synthetic data only
- Precision and recall for spikes are modest at the default threshold
- Not clinically validated, so not for medical decisions
- Single adverse event: glucose spike

Next steps

- Validate on anonymized clinical data (e.g. MIMIC-IV) and open wearable datasets
- Class weighting, threshold tuning and richer features
- Real-time streaming ingestion from devices
- More adverse events, such as cardiovascular anomalies



Synthetic data only. No real patient data.

Thank you

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Code, data generators, dashboard and docs are in the GitHub repository. Released under the MIT License.